Matthew Berger

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Senior Staff Rust engineer with 10+ years writing Rust (including pre-1.0). Founding engineer at Hyphen; built food assembly robotics controls suite from zero to production in 22 months, anchoring $50M+ in combined investment from Chipotle and Cava.

Work Experience

Hyphen (Culinary Robotics) — Remote
Staff → Senior Staff Software Engineer (Founding Engineer) — September 2022 - May 2026

- Sole architect of the production food assembly robotics controls suite; merged **1,000+ pull requests** over 4 years across firmware, controls, IPC, and cloud infrastructure
- Engineered full technology stack from bare-metal firmware (RP2040/Embassy) to distributed robotics control system, custom Yocto Linux distribution, and AWS cloud
- Built core async message broker coordinating distributed processes on industrial PC with direct networking to embedded boards, handling **1M+ msgs/sec** with **<1ms latency**, zero message loss over **3 years** of constant use
- Built Hyphen Explorer (Bevy + egui) for real-time IPC visualization, debugging, and system control, achieving **100% adoption** across all Hyphen engineers
- Shipped Hyphen Studio, a Dioxus desktop application enabling operators to configure makeline layouts and parameters directly
- Replaced TypeScript/GraphQL tablet portal with a pure Rust Leptos rewrite connecting directly to the IPC broker, eliminating an entire backend tier
- Mentored team of web developers into Rust systems programmers through architectural review, pair programming, and codified patterns

Hyphen (Culinary Robotics) — Remote
Senior Software Engineer (Founding Engineer) — July 2021 - September 2022

- Diagnosed critical scaling limitations in legacy PLC system preventing growth beyond prototype
- Led technical pivot from $500K PLC system to embedded Rust architecture
- Pioneered RP2040 firmware with Embassy-rs, creating first Rust TMC5160 motor and AS5048A encoder drivers proving Rust could meet real-time control loop requirements

Sierra Nevada Corporation — Englewood, CO
Software Engineer III — May 2020 - July 2021

- Developed C++/Rust aerospace imaging system (5GB/sec) and simulator saving 3 months on delivery timeline

Scientific Games — Reno, NV
Software Engineer — July 2019 - May 2020

- Shipped performance and gameplay defect fixes in a Unity-based casino platform

Hamilton Company Reno, NV
Software Engineer January 2018 - July 2019
Software Engineering Intern October 2014 - December 2017

- Built safety-critical software for liquid-handling medical robots
- Automated gravimetric analysis QA process, saving **1+ week per robot** across production fleet
- Reduced calibration development from **2 months to 2 weeks** through reusable plugin framework
- Architected diagnostic application for largest OEM customer Illumina (**$2M contract**)

Skills

Languages: Rust, C++, C, C#, TypeScript, Python
Systems: Embedded (RP2040, Embassy), real-time control, async (Tokio), IPC, message brokers, Yocto Linux, distributed systems
Web/UI: Leptos, Dioxus, egui, WebAssembly, wgpu, Vulkan, OpenGL
Cloud/Infra: AWS, Pulumi, Greengrass IoT, Docker, GitHub Actions

Open Source

- **Nightshade**: data-oriented Rust game engine featuring a custom render graph, built-in editor, rapier3d physics, kira audio, gltf asset loading, procedural terrain, and Steam integration; runs natively on DX12/Metal/Vulkan and in-browser via WebGPU (live demo: **matthewberger.dev/nightshade**)
- **freecs** (**48k+ downloads** on crates.io): archetype-based ECS in Rust with zero unsafe code, Rayon-parallel system execution, sparse-set tags, deferred command buffers, and watermark change detection; the entire ECS is generated at compile time via a declarative macro. Design documented in a 3-part series at **matthewberger.dev/articles**
- **enum2 macro family** (enum2str, enum2contract, enum2egui, enum2pos, enum2repr): derive macros for enum-driven design patterns with **230k+ total downloads** on crates.io
- **wgpu-example**: minimal Rust + wgpu + egui template showing how to build graphics that run everywhere; native desktop, WebGPU and WebGL via WebAssembly, Android, Steam Deck, and OpenXR VR with hand tracking
- **cameras** (new in 2026): cross-platform Rust camera library with data-oriented design (explicit format negotiation, push-based frame delivery, zero trait objects); ships with Dioxus and egui integrations

Education

University of Nevada, Reno BS Computer Science & Engineering, Minor in Mathematics, 2017